

VITAMIN B₉ (FOLATE)

Etiology and Pathogenesis

Folate is present in a wide variety of foods, particularly liver, wheat bran, other grains, leafy green vegetables, and dried beans. The active coenzyme form, tetrahydrofolate, plays a critical role in single-carbon transfer reactions involved in amino acid, purine, and pyrimidine metabolism.

Folate deficiency can result from: - **Malabsorptive states:** such as celiac disease, chronic diarrhea, or post-total gastrectomy. - **Antifolate medications:** including methotrexate, trimethoprim, oral contraceptives, and pyrimethamine. - **Antiepileptic drugs:** phenobarbital and phenytoin induce hepatic microsomal enzymes, accelerating folate metabolism and depleting stores. - **Poor dietary intake:** commonly seen in alcoholics. - **In children:** excessive boiling of milk or consumption of goat's milk—both associated with low folate bioavailability compared to human or cow's milk.

Clinical Findings

The primary manifestation is **hematologic:** - Hypersegmented neutrophils - Macrocytosis - Megaloblastic anemia - Neutropenia - Thrombocytopenia

Other systemic features include diarrhea and irritability.

Unlike vitamin B₁₂ deficiency, **folate deficiency does not cause neurologic symptoms.**

Mucocutaneous findings may include: - Glossitis with atrophy of filiform papillae - Angular cheilitis - Mucosal and perirectal ulcerations - Perineal seborrheic dermatitis - Diffuse brown hyperpigmentation, especially in palmar creases and flexural areas

Laboratory Testing

- Macrocytic anemia with megaloblastic changes and neutrophil hypersegmentation on peripheral smear is suggestive.
- **Serum folic acid levels** confirm the diagnosis.

Treatment

- **Folic acid supplementation** (1–5 mg/day) is typically curative.
- Discontinue any implicated antifolate medications if possible.
- **Crucially, rule out vitamin B₁₂ deficiency before initiating folate therapy.**

If B₁₂ deficiency is present but untreated, folate may correct the anemia while allowing neurologic deterioration to progress unchecked.

VITAMIN B₁₂ (COBALAMIN)

Etiology and Pathogenesis

Vitamin B₁₂ serves as a coenzyme in two key metabolic pathways: 1. **Methylcobalamin:** acts as a cofactor for methionine synthase, which converts homocysteine to methionine—essential for DNA, protein, and lipid synthesis. 2. **5'-Adenosylcobalamin:** required by methylmalonyl-CoA mutase to convert methylmalonic acid to succinyl-CoA, a step in fat and carbohydrate metabolism.

Dietary sources are primarily animal-based: liver, eggs, milk, beef, and organ meats.

Absorption process: - Gastric acid releases B₁₂ from dietary proteins. - It then binds to intrinsic factor in the duodenum. - The intrinsic factor–B₁₂ complex is absorbed via specific receptors in the terminal ileum. - Within enterocytes, B₁₂ detaches from intrinsic factor and enters the bloodstream bound to transcobalamin II for tissue delivery. - A small amount (1–5%) of free B₁₂ is absorbed by passive diffusion along the intestinal wall.

The body stores significant amounts of vitamin B₁₂, primarily in the liver, so deficiency typically develops slowly over 3–6 years.

Epidemiology

Causes of vitamin B₁₂ deficiency fall into three main categories: - **Inadequate intake** - **Malabsorption** - **Other causes**